

Lukas Helmig

PhD Student in Theoretical Physics

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📍 Osnabrück, Germany

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Education

PhD	Osnabrück University , Physics	Osnabrück, Germany
	• Investigating equilibration in disordered systems.	2025 – 2028
Master of Science	University of Cologne , Physics	Cologne, Germany
	• Top of the year.	2022 – 2025
	• Specialized in solid state theory and quantum information.	
	• Thesis on "Restricted Boltzmann Machines and Hopfield Networks".	
Erasmus exchange	Ecole Normale Supérieure de Paris , Physics	Paris, France
	• Participated in the Erasmus exchange program, focusing on advanced topics in theoretical physics.	2023 – 2024
Bachelor of Science	Osnabrück University , Physics	Osnabrück, Germany
	• Top of the year. Finished with honors.	2018 – 2021
	• Thesis on "Influence of interactions on the efficiency at maximal power of a microscopic thermodynamic machine".	

Experience

Faculty of Physics, Osnabrück University , Research Assistant	Osnabrück, Germany
Worked on the VRnano project, a physical development environment for better control of laboratory experiments.	2021 – 2023 2 years
Faculty of Mathematics, Osnabrück University , Teaching Assistant	Osnabrück, Germany
Tutorial lectures and exam grading.	2019 – 2021 2 years

Awards

Honors branch	University of Cologne
Awarded for outstanding academic achievement.	2022
Niedersachsen Scholarship	Niedersachsen Ministry of Science and Culture
Awarded for outstanding academic achievement.	2021
Deutschlandstipendium	Osnabrück University
Awarded for outstanding academic achievement (2019-2021).	2019

Teaching

Theoretical Physics III	Osnabrück University, Germany
The course covers advanced topics in classical mechanics and electrodynamics. Held weekly exercise classes and provided feedback to students.	2026 – 2026
Theoretical Physics IV	Osnabrück University, Germany
The course covers advanced topics in quantum mechanics and statistical mechanics. Held weekly exercise classes and provided feedback to students.	2025 – 2026

Outreach

Volunteer at DOPPLERS

DOPPLERS is a yearly theoretical physics competition held at different german universities and supported by the German Physical Society (DPG). I organized the academic program including two speakers and held a workshop on decision-making myself. Also, I sought and acquired sponsoring for the catering. On the weekend of the event itself, I was volunteering and helping with the event in general.

German Physical Society
(DPG), Osnabrück, Germany
2026